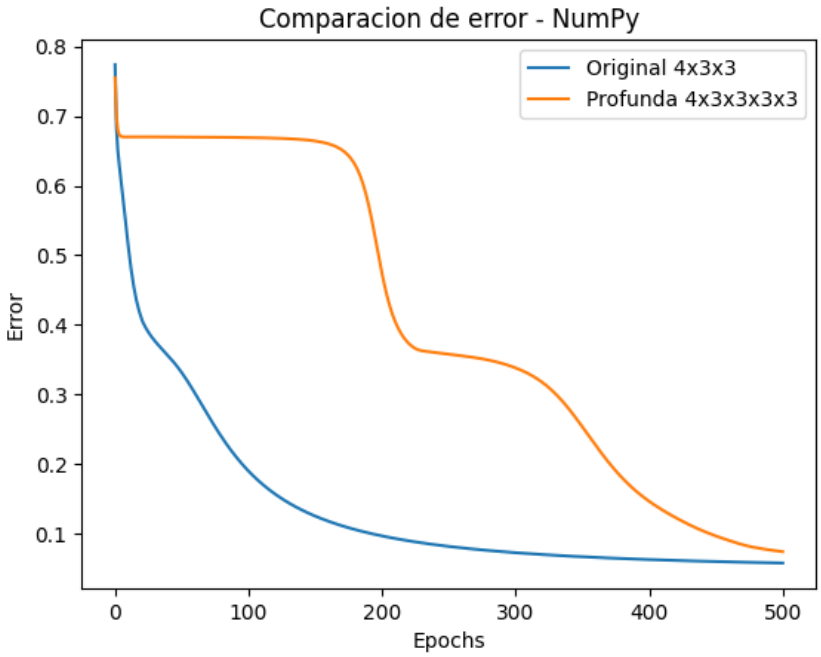
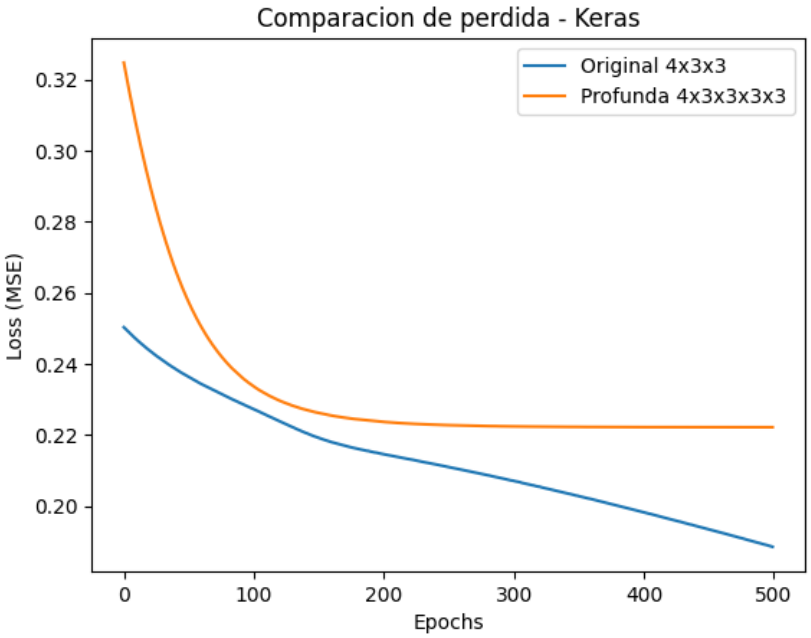
Capturas de evidencia

Gráficas de error/pérdida notebook 04 Multilayer perceptrón



Gráficas de error/pérdida notebook 05 Keras – multilayer perceptrón – iris



Model.summary() Keras – Original

```
model = keras.Sequential(
  [
    layers.Dense(3, activation="sigmoid", name="layer1", input_shape=(4,)),
    layers.Dense(3, activation="sigmoid", name="layer2"),
  ]
)

model.summary()
```

```
/usr/local/lib/python3.13/dist-packages/keras/src/layers/core/dense.py:106: UserWarning
  super().__init__(activity_regularizer=activity_regularizer, **kwargs)
Model: "sequential"
```

Layer (type)	Output Shape	Param #
layer1 (Dense)	(None, 3)	15
layer2 (Dense)	(None, 3)	12

```
Total params: 27 (108.00 B)
Trainable params: 27 (108.00 B)
Non-trainable params: 0 (0.00 B)
```

Model.summary() Keras – Profundo

```
model_deep = keras.Sequential(
  [
    layers.Dense(3, activation="sigmoid", name="layer1", input_shape=(4,)),
    layers.Dense(3, activation="sigmoid", name="layer2"),
    layers.Dense(3, activation="sigmoid", name="layer3"),
    layers.Dense(3, activation="sigmoid", name="layer4"),
  ]
)

model_deep.summary()
```

```
/usr/local/lib/python3.13/dist-packages/keras/src/layers/core/dense.py:106: UserWarning:
  super().__init__(activity_regularizer=activity_regularizer, **kwargs)
Model: "sequential_1"
```

Layer (type)	Output Shape	Param #
layer1 (Dense)	(None, 3)	15
layer2 (Dense)	(None, 3)	12
layer3 (Dense)	(None, 3)	12
layer4 (Dense)	(None, 3)	12

```
Total params: 51 (204.00 B)
Trainable params: 51 (204.00 B)
Non-trainable params: 0 (0.00 B)
```